

PVsyst - Simulation report

Standalone system

Project: Rio_Hacha

Variant: Rio_HACHA_Metodo2

Standalone system with batteries

System power: 10.20 kWp

Rio Hacha - Colombia

**PVsyst V8.0.19**

VCO, Simulation date:
20/12/25 19:08
with V8.0.19

Project summary**Geographical Site**

Rio Hacha
Colombia

Situation

Latitude 11.53 °(N)
Longitude -72.91 °(W)
Altitude 9 m
Time zone UTC-5

Project settings

Albedo 0.20

Weather data

Rio Hacha
PVGIS api TMY

System summary**Standalone system**

Orientation #1
Horizontal plane

Standalone system with batteries

Orientation #2
Fixed plane
Tilt/Azimuth 20 / 0 °

User's needs

Daily household consumers
Constant over the year
Average 48 kWh/Day

System information**PV Array**

Nb. of modules 340 units
Pnom total 10.20 kWp

Battery pack

Technology Lithium-ion, LFP
Nb. of units 4020 units
Voltage 48 V
Capacity 831 Ah

Results summary

Useful energy from solar	16566 kWh/year	Specific production	1624 kWh/kWp/year	Perf. Ratio PR	80.75 %
Missing Energy	962 kWh/year	Available solar energy	17917 kWh/year	Solar Fraction SF	94.51 %
Excess (unused)	835 kWh/year				

Table of contents

Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
Detailed User's needs	5
Main results	6
Loss diagram	7
Predef. graphs	8

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General parameters**Standalone system**

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Average

48 kWh/Day

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Fixed plane

Tilt/Azimuth

20 / 0 °

Models used

Transposition

Diffuse

Circumsolar

Perez

Imported

separate

PV Array Characteristics**PV module**

Manufacturer

Generic

Model

Poly 30 Wp 36 cells

(Original PVsyst database)

Unit Nom. Power

30 Wp

Number of PV modules

340 units

Nominal (STC)

10.20 kWp

Modules

85 string x 4 In series

At operating cond. (50°C)

Pmpp

9.98 kWp

U mpp

56 V

I mpp

179 A

Total PV power

Nominal (STC)

10 kWp

Total

340 modules

Module area

79.6 m²**Controller**

Universal controller

Technology

MPPT converter

Temp coeff.

-5.0 mV/°C/Elem.

Converter

Maxi and EURO efficiencies

97.0 / 95.0 %

Battery Management control

Threshold commands as

SOC calculation

Charging

SOC = 0.96 / 0.80

Discharging

SOC = 0.10 / 0.35

Battery Storage**Battery**

Manufacturer

Generic

Model

Cell 26650 3Ah 3.2V

Technology

Lithium-ion, LFP

Nb. of units

268 in parallel x 15 in series

Discharging min. SOC

10.0 %

Stored energy

34.5 kWh

Battery Pack Characteristics

Voltage

48 V

Nominal Capacity

831 Ah (C10)

Temperature

Fixed 20 °C

Array losses**Thermal Loss factor**

Module temperature according to irradiance

Uc (const)

20.0 W/m²K

Uv (wind)

0.0 W/m²K/m/s**DC wiring losses**

Global array res.

5.2 mΩ

Loss Fraction

1.50 % at STC

Serie Diode Loss

Voltage drop

0.7 V

Loss Fraction

1.1 % at STC

Module Quality Loss

Loss Fraction

-0.75 %

Module mismatch losses

Loss Fraction

2.00 % at MPP

Strings Mismatch loss

Loss Fraction

0.15 %



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Array losses

IAM loss factor

Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.402	0.000



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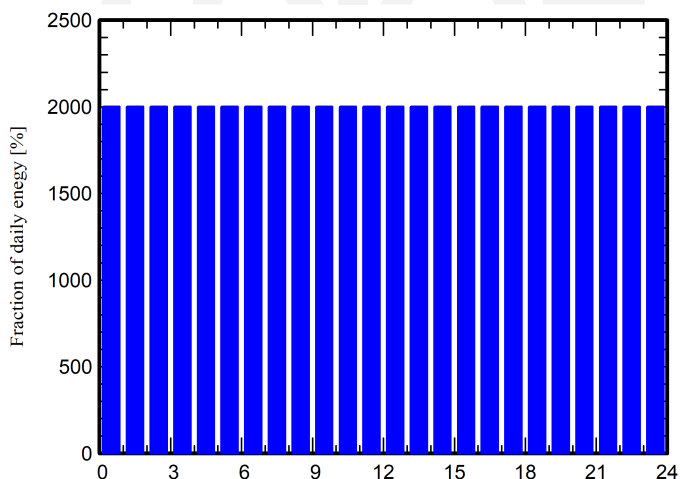
Detailed User's needs

Daily household consumers, Constant over the year, average = 48 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Otros usos	1	2000 tot	24.0	48000
Consumidores en espera			24.0	24
Total daily energy				48024

Hourly distribution





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Main results

System Production

Useful energy from solar 16566 kWh/year
Available solar energy 17917 kWh/year
Excess (unused) 835 kWh/year

Perf. Ratio PR 80.75 %
Solar Fraction SF 94.51 %

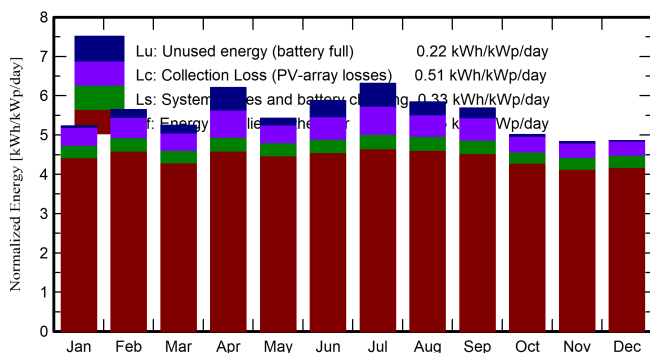
Loss of Load

Time Fraction 5.5 %
Missing Energy 962 kWh/year

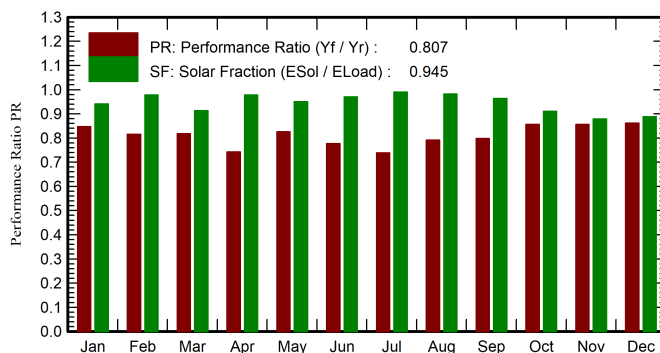
Battery ageing (State of Wear)

Cycles SOW 95.7 %
Static SOW 90.0 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m ²	kWh/m ²	kWh	kWh	kWh	kWh	kWh	ratio
January	162.2	156.7	1452	7.2	87.7	1401	1489	0.941
February	158.1	153.6	1410	53.3	29.6	1315	1345	0.978
March	162.8	159.0	1461	60.4	129.6	1359	1489	0.913
April	186.1	181.6	1624	171.6	32.0	1409	1441	0.978
May	168.1	164.0	1512	50.5	73.1	1416	1489	0.951
June	176.3	172.1	1563	123.3	43.7	1397	1441	0.970
July	195.8	191.2	1706	180.3	14.3	1474	1489	0.990
August	181.0	177.0	1609	100.2	27.3	1461	1489	0.982
September	170.5	166.7	1502	70.8	52.2	1389	1441	0.964
October	155.4	151.2	1401	11.1	132.2	1357	1489	0.911
November	145.0	140.5	1310	5.9	175.2	1266	1441	0.878
December	150.6	145.1	1368	0.0	165.5	1323	1489	0.889
Year	2011.8	1958.6	17917	834.7	962.3	16566	17529	0.945

Legends

GlobHor Global horizontal irradiation
GlobEff Effective Global, corr. for IAM and shadings
E_Avail Available Solar Energy
EUnused Unused energy (battery full)
E_Miss Missing energy

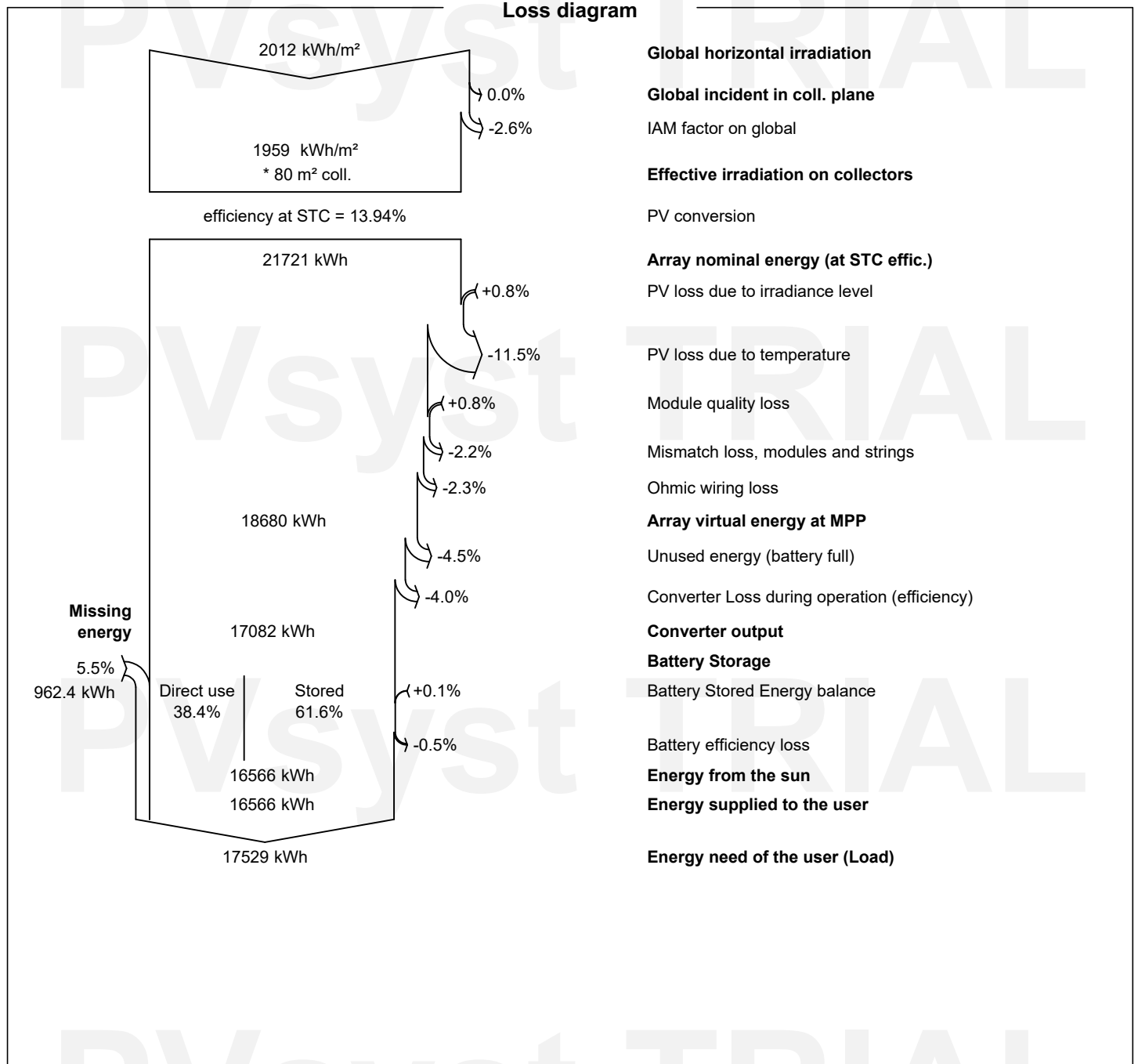
E_User Energy supplied to the user
E_Load Energy need of the user (Load)
SolFrac Solar fraction (EUsed / ELoad)



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Loss diagram





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Predef. graphs
Daily Input/Output diagram

